Tallinn Apartment Livability Score

Methodology summary

7,842	4	5
apartment buildings scored	proximity factors, equal weighted	livability categories

1 Data sources

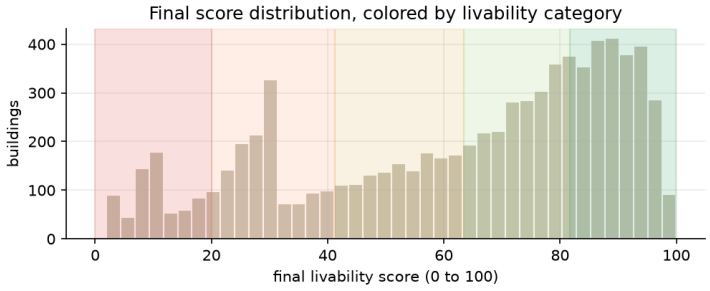
Source	Provides	Count in Tallinn
ETAK	Building footprints	7,842 buildings
EHR	Energy class (energiamärgis), address	3,807 with energy data
ETAK	Green and blue space polygons	3,128 polygons
OpenStreetMap	Street network for walking routes	277,733 nodes
Ühistranspordiregister	Transit stops, GTFS feed	1,319 stops
OpenStreetMap (Overpass)	Grocery and gym locations	236 / 56

ETAK (Eesti topograafia andmekogu, the Estonian Topographic Database) and EHR (Ehitisregister, the Estonian Building Register) are maintained by the Estonian Land and Spatial Development Board (Maa- ja Ruumiamet); building footprints and energy class are delivered together through its etak_ehr_hooned WFS service, which joins the two registers upstream. Ühistranspordiregister is the Estonian Public Transport Register. Buildings are limited to apartment buildings with 3 or more units and clipped to the Tallinn city boundary; the transit and amenity counts above are also clipped to that boundary.

2 How the score is generated

- Network distance: the shortest walking route on the real street network to the nearest transit stop, grocery, gym, and green or blue space.
- Distance decay: closer amenities score higher, using a smooth curve rather than a hard cutoff. A location right at the typical walking threshold for that amenity type scores exactly halfway.
- Normalize: each factor is rescaled to a common 0 to 100 range.
- Combine: the four scores are combined using a method where a very poor result on one factor cannot be fully offset by strong results on the others.

Note on energy score: Estonia's building energy certificate (A to H) and renovation history were both considered as scoring factors. Neither is used in the final score. Only about half of buildings have a certificate on file, and no reliable renovation date exists for any building. Energy class is still shown on the map for reference, it just does not affect the score or category.

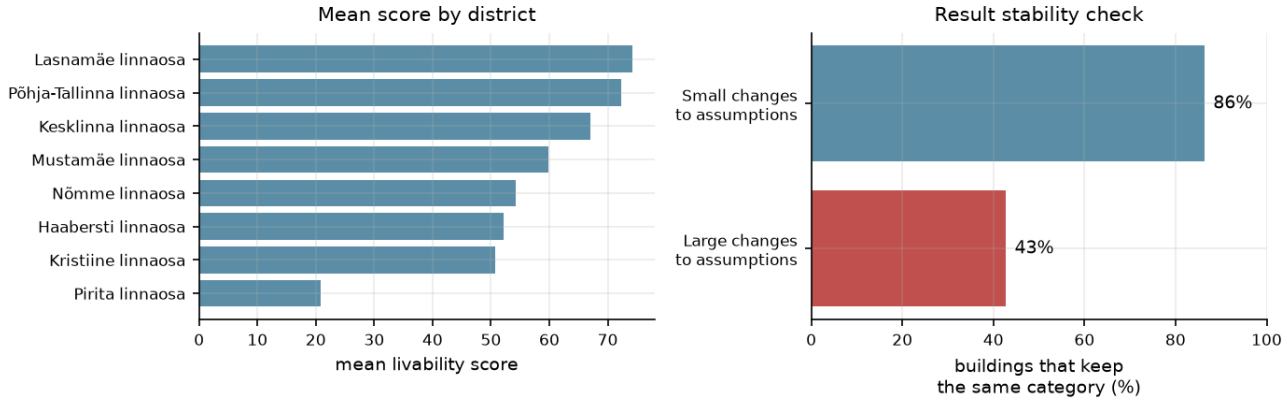


Final score distribution across all buildings, shaded by the resulting category.

3 Validation

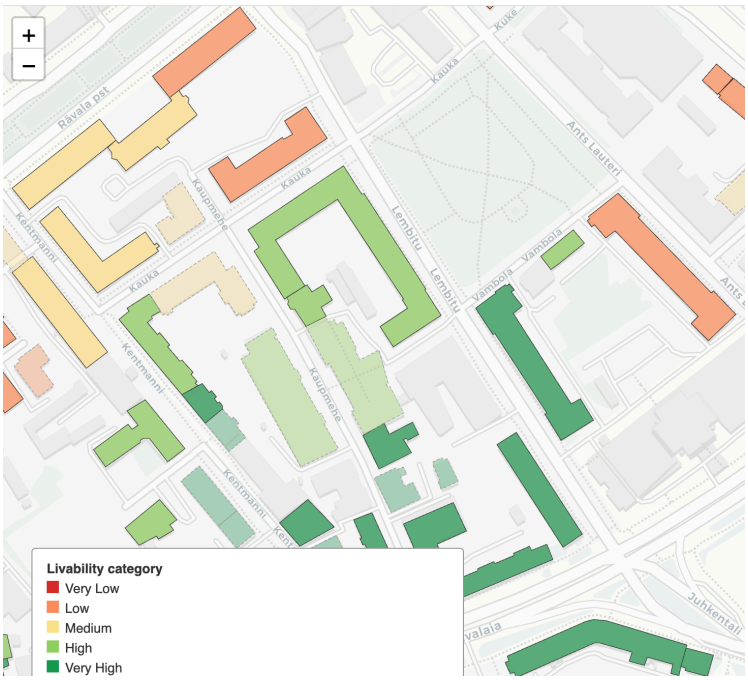
After the score was calculated, it was checked three ways: manually reviewing the highest and lowest scoring buildings, comparing average scores across districts against what is generally known about the city, and testing how much the results change if the underlying assumptions change.

- Manual review caught a real error: about 2 percent of buildings were being scored as if they had no nearby amenities at all, because of a gap in the street network data. This has been fixed, and every building now routes correctly.
- District averages line up with expectations: busier, well connected districts such as Lasnamäe and Põhja-Tallinna score highest, while the quieter, more spread out Pirita district scores lowest.
- Sensitivity testing: using a Monte Carlo simulation, the score was recalculated 300 times with varied assumptions. Small, reasonable changes barely move the results. Large, unrealistic changes move them a lot, which is expected and is part of why the simpler, equal weighted approach was chosen.



Left: average score by district. Right: share of buildings that keep the same category when the scoring assumptions are changed a little versus a lot.

4 Example: the map in practice



A sample area of central Tallinn, buildings colored by livability category. A dashed outline means no energy certificate is on file for that building; this is shown for reference only and does not change its score.